

Solution to Problem 8

2024 Rudolf Ortvy Competition in Physics

Mohammadreza Shokouhi

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Abstract

This document presents a theoretical analysis of the total work required to quasistatically lift a uniform massive rope from a rough horizontal surface. Assuming continuous mechanical equilibrium, the geometric profile of the suspended rope is modeled as a catenary. A closed-form expression for the total work done is derived as a function of the coefficient of kinetic friction, μ . The physical validity of this analytical model is subsequently corroborated by evaluating the asymptotic limits of a frictionless surface ($\mu \rightarrow 0$) and a perfectly rough surface ($\mu \rightarrow \infty$).

1 Total Work Done

Let $\lambda = m/L$ denote the uniform linear mass density of the rope. Assuming the rope is lifted quasistatically, the system can be considered to remain in mechanical equilibrium at all times. Let T represent the applied force lifting the rope from the ground. This force is not strictly vertical, as the horizontal tension along the suspended segment of a catenary must remain constant. At any given instant, let s denote the length of the rope resting on the ground, leaving a suspended length of $L - s$. Furthermore, let x and y define the horizontal distance from the grounded segment to the lifting axis and the vertical height of the applied force, respectively. The configuration of the system during this process is illustrated in Fig. 1b.

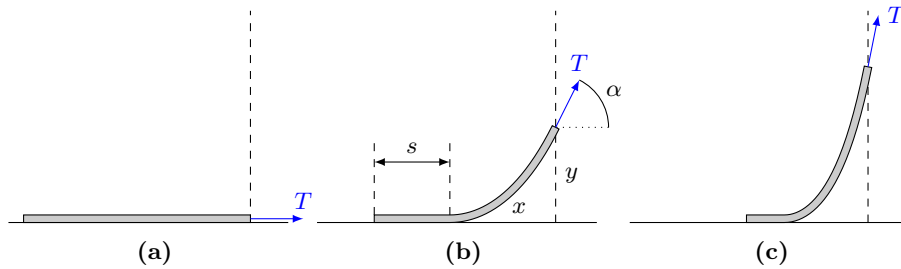


Figure 1 Free-body configuration of a massive rope being lifted quasistatically, ensuring continuous mechanical equilibrium.

To determine the total work done, establishing the geometric profile of the suspended segment is essential. Based on standard catenary mechanics (e.g.,

see [1], Problem 2.8), the profile and slope are given by

$$y = a \cosh\left(\frac{x}{a}\right) - a, \quad \tan(\alpha) = \sinh\left(\frac{x}{a}\right), \quad (1)$$

where $a \equiv T_0/(\lambda g)$ and T_0 denotes the constant horizontal tension within the rope. These relations hold for any point along the suspended segment when the origin is located at the detachment point on the ground. At this junction, the horizontal tension must balance the kinetic friction acting on the grounded segment; thus, $T_0 = \mu \lambda s g$, which simplifies to $a = \mu s$. For the boundary condition at the lifted end, force equilibrium dictates:

$$T \sin(\alpha) = \lambda(L - s)g, \quad T \cos(\alpha) = T_0 \rightarrow \tan(\alpha) = \frac{L - s}{\mu s}.$$

Substituting this slope into (1) and eliminating x yields the vertical displacement of the hand as a function of s :

$$y = \sqrt{\mu^2 s^2 + (L - s)^2} - \mu s. \quad (2)$$

Because only the vertical component of the applied force contributes to the mechanical work, the total work required to lift the entire rope evaluates to

$$\begin{aligned} W &= \int_0^L \lambda(L - s)g \, dy \\ &= \lambda g \int_0^L \left[\mu - \frac{(1 + \mu^2)s - L}{\sqrt{\mu^2 s^2 + (L - s)^2}} \right] (L - s) \, ds \quad (\text{using (2)}) \\ &= mgL \int_0^1 \left[\mu + \frac{1 - (1 + \mu^2)u}{\sqrt{1 - 2u + (1 + \mu^2)u^2}} \right] (1 - u) \, du. \quad (u = s/L) \end{aligned}$$

Evaluating this integral provides the dimensionless work as a function of the friction coefficient, μ :

$$\frac{W(\mu)}{mgL} = \frac{(1 + \mu + 2\mu^2) \sqrt{1 + \mu^2} - \mu^2 [\sinh^{-1}(\mu) + \sinh^{-1}(1/\mu)]}{2(1 + \mu^2)^{3/2}}. \quad (3)$$

For a specified friction coefficient of $\mu = 0.3$, the required work is approximately:

$$W(\mu = 0.3) \approx 0.591mgL.$$

The dependence of the required work on the coefficient of friction, μ , is illustrated graphically in Fig. 2.

2 Limiting Cases $\mu \rightarrow 0$ and $\mu \rightarrow \infty$

To verify the validity of (3), it is instructive to examine two asymptotic limits. The first is the limit of a frictionless surface ($\mu \rightarrow 0$). In this regime, the expression simplifies to:

$$\lim_{\mu \rightarrow 0} W(\mu) = \frac{1}{2}mgL.$$

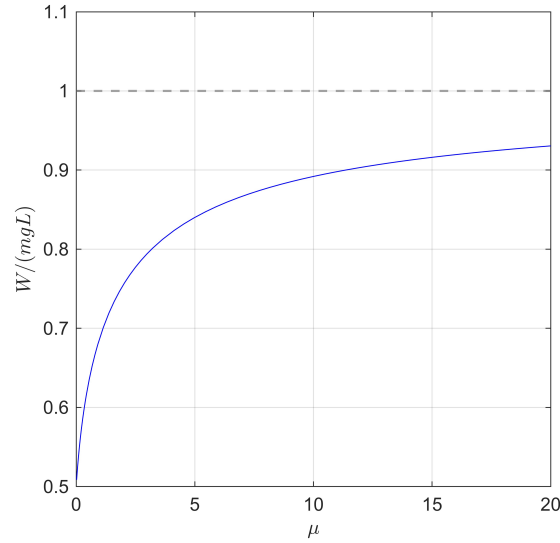


Figure 2 Total required work to lift the rope, normalized by mgL , plotted as a function of the coefficient of friction, μ .

This result is strictly consistent with the principle of energy conservation. In the absence of dissipative friction, the work performed by the external force is entirely converted into the gravitational potential energy of the rope. Given that the center of mass is elevated from the ground to a final height of $L/2$, the corresponding change in potential energy is precisely $mgL/2$, confirming the derived mathematical limit.

The second scenario of interest is the limit of a perfectly rough surface ($\mu \rightarrow \infty$). Evaluating (3) for this case yields:

$$\lim_{\mu \rightarrow \infty} W(\mu) = mgL.$$

This result can be rationalized through physical arguments: as the coefficient of friction approaches infinity, horizontal dragging of the rope becomes physically prohibited. Consequently, the rope can only be lifted vertically without any horizontal displacement, implying that the horizontal tension vanishes identically. Under these conditions, the normal force exerted on the grounded segment is zero throughout the lifting process. The applied lifting force must therefore continuously support the entire weight of the suspended portion, effectively treating the system as a vertical lift, which correctly yields a total work of mgL .

References

- ¹D. Morin, *Introduction to classical mechanics: with problems and solutions* (Cambridge University Press, Cambridge, 2008).